

Synthesis of chiral amino acids with metal ion chelating side chains from L-serine using Negishi cross-coupling reaction

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Abstract—The scope of the Negishi cross-coupling reaction of organozinc compounds derived from chiral amino acids was extended to electron rich iodoanilines and iodobenzylamines as coupling reagents. The protocol allows the direct modification of serine into phenylalanine derivatives bearing metal ion chelating ligands in their side chain, such as amino esters **6** and **7**. The preparation of metal complex labeled peptides, the construction of synthetic receptors and hybrid materials are potential applications of the modified chiral amino acids.

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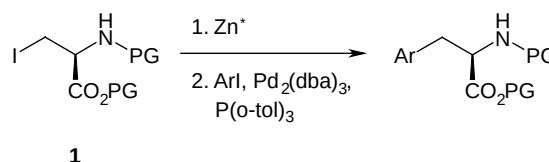
1. Introduction

The function of peptides and proteins is strongly correlated with their secondary and tertiary structure,¹ which is induced by side chain interactions and determined through the amino acid sequence. Non-proteinogenic amino acids^{2–4} and amino acid mimics are widely used to define,⁵ modulate⁶ or switch⁷ the structure of peptides, which may result in changes of their biological properties. One concept⁸ to control a peptide structure is the inter- or intramolecular formation of a metal complex, whereby parts of the peptide function as donor ligands. This requires modified amino acids bearing a strong metal ion chelating ligand for stable complex formation for incorporation into a peptide sequence.⁹ Different research groups used alkylation reactions of lysine or diaminopropionic acid to generate imino diacetic acid (IDA)¹⁰ or bis-(2-picolyl)-amine (BPA)¹¹ chelates directly on protein surfaces.

As this modification of amino acid side chains is limited to a certain kind of chelates a more general solution to transform natural amino acids into metal ion chelating amino acids is desired.

Jackson already showed that palladium-catalyzed coupling of the organozinc compound derived from **1**¹² with aryl iodides is a feasible route to derivatives of phenylalanine (Scheme 1).¹³ We extend the scope of this reaction to

phenylalanine derivatives bearing *p*-NH₂ (**4**), *p*-NH-Boc (**5**), and *p*-Me (**3**) aryl substituents and report the efficient synthesis of the non-natural chiral amino acids **6** and **7** having protected IDA and cyclen metal ion chelating ligands in their side chain.



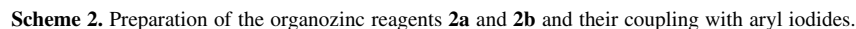
Scheme 1. Palladium-catalyzed coupling of aryl iodides with **1**.

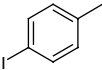
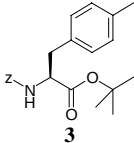
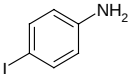
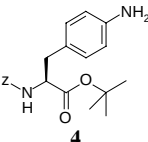
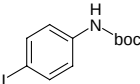
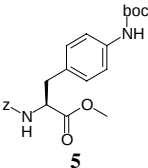
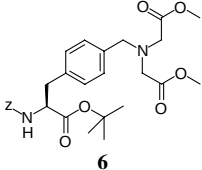
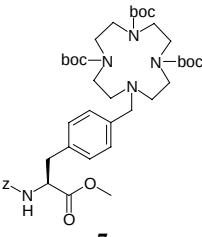
2. Results and discussion

So far, no Negishi coupling¹⁴ reaction has been reported for alkylorganozinc compounds with *p*-iodoaniline derivatives as reactants.¹⁵ Previous preparations of the unnatural amino acid H-*p*-NH₂-Phe-OH and protected derivatives of it used either the reduction of *p*-nitrophenyl alanine¹⁶ or Staudinger reactions¹⁷ of *p*-azidophenyl alanine. The required iodo precursors **1a**^{18,19} and **1b**²⁰ were synthesized from serine. The corresponding organozinc reagents **2a** and **2b** were obtained by direct insertion of freshly prepared Rieke zinc.²¹ The alternative method of zinc dust activation by Me₃SiCl in DMF was in our hands not as efficient as the use of Rieke zinc.²² The disappearance of the black zinc indicates the complete conversion by insertion reaction (Scheme 2).²³

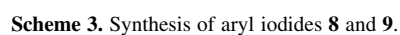
Keywords: Amino acids; Metal chelates; Organozinc compounds; C–C bond formation; Negishi coupling.

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Entry	Iodoalanine	Ar-I	Solvent	Yield (%)	Product
1	2a		DME	66	
2	2a		HMPT/THF = 1:1	73	
3	2b		HMPT/THF = 1:1	75	
4	2a	9	HMPT/THF = 1:1	65	
5	2b	8	HMPT/THF = 1:1	86	

product (Table 1, entry 1) using DME as solvent with 0.1 equiv CuBr·SME₂,¹⁸ the more electron rich *p*-iodoaniline requires a solvent mixture of HMPT and THF (1:1) to give 73% of *Z*-*p*-NH₂-Phe-O^tBu **4** (Table 1, entry 2). To



apply the method to the synthesis of metal chelating chiral amino acids, aryl iodide substituted protected metal chelates were prepared starting from *p*-iodobenzyl bromide. Methyl imino diacetate and trifold Boc-protected cyclen²⁴ were alkylated in good yield giving aryl iodides **8** and **9** (Scheme 3).

Applying the same conditions as used for the coupling of *p*-iodoaniline, IDA derivative **6** was obtained in moderate 65% yield (Table 1, entry 4). Negishi coupling of aryl iodide **8** with iodoalanine **2a** gave cyclen-modified amino acid **7** in good yield of 86% (Table 1, entry 5).

3. Conclusion

We have extended the scope of the Negishi cross-coupling reaction of organozinc compounds derived from chiral amino acids to electron rich iodoanilines and iodobenzylamines as coupling reagents. This allows now the direct modification of serine into phenylalanine derivatives **4** and **5** and amino esters **6** and **7**, bearing metal ion chelating ligands in their side chain. Such and similar modified chiral amino acids are useful for the preparation of peptides labeled with metal complexes as binding sites or probes, for the construction of synthetic receptors and hybrid materials having a peptide backbone and metal complex functionality.

4. Experimental

4.1. General procedure for the generation of organozinc reagents **2a** and **2b**

Rieke zinc was prepared from ZnCl₂ using a catalytic amount of naphthalene: to a Schlenk tube charged with finely divided lithium (1% Na) (3.0 mmol), naphthalene (0.3 mmol) and 2 mL of anhydrous THF (distilled from benzophenone–sodium ketyl under N₂) or DME (in case of *Z-p*-Me-Phe-O^tBu **3**) under argon a solution of anhydrous zinc chloride [1.5 mmol in 2 mL of THF or DME (in case of *Z-p*-Me-Phe-O^tBu **3**)] was transferred via syringe. The mixture was stirred vigorously until all lithium was consumed (0.5 h). To the Rieke zinc suspension under argon a solution of iodoalanine (1.28 mmol) in 3 mL of DME (in case of *Z-p*-Me-Phe-O^tBu **3**) or THF was added via syringe. The exothermic reaction was completed in 5 min (the end of reaction is indicated when the black zinc disappeared). The mixture was stirred for an additional 0.5 h.

4.1.1. Tri-*tert*-butyl-10-(*p*-iodo-benzyl)-1,4,7,10-tetraaza cyclododecane-1,4,7-tricarboxylate (8**).** *p*-Iodobenzyl bromide (0.5 g, 1.7 mmol) was dissolved in 35 mL of MeCN. Trifold-Boc protected cyclen (1 g, 2.2 mmol) and K₂CO₃ (1.21 g, 8.8 mmol) were added, and after completion of the reaction the reaction mixture was filtered and concentrated. The crude product was purified by chromatography on silica gel (hexanes/dichloromethane=5:1 → AcOEt) giving **8** (1.09 g, 93%) as a colorless solid, mp=75–77 °C. *R*_f (AcOEt)=0.84; ¹H NMR (CDCl₃, 300 MHz): δ=1.25–1.56 (m, 27H), 2.39–2.78 (m, 4H), 3.11–3.73 (m, 14H), 7.00 (d, *J*=8.2 Hz, 2H), 7.61 (d, *J*=8.2 Hz, 2H); ¹³C

NMR (CDCl₃, 75 MHz): δ=28.5, 28.7, 47.9, 50.0, 54.8, 55.5, 56.7, 79.5, 92.8, 132.2, 136.7, 137.2, 155.4, 155.7, 156.1; LR-MS (ESI/DCM/MeOH+10 mmol/L NH₄OAc): *m/z*=689 [(M+H)⁺].

4.1.2. Methyl [(4-iodo-benzyl)-methoxycarbonylmethylamino]-acetate (9**).** Dimethyl iminodiacetate hydrochloride (0.87 g, 4.4 mmol) and NaH (60% suspension) (0.18 g, 4.4 mmol) were dissolved in 35 mL of MeCN and stirred for 5 min. K₂CO₃ (2.42 g, 17.5 mmol) and *p*-iodobenzyl bromide (1 g, 3.4 mmol) were added to this suspension. After completion of the reaction, the mixture was filtered and concentrated. The crude product was purified by chromatography on silica gel (hexanes/dichloromethane=5:1 → AcOEt) yielding **9** (1.49 g, 90%) as a colorless oil. *R*_f (hexanes/dichloromethane=5:1)=0.44; ¹H NMR (CDCl₃, 300 MHz): δ=3.52 (s, 4H), 3.68 (s, 6H), 3.84 (s, 2H), 7.13 (d, *J*=8.2 Hz, 2H), 7.62 (d, *J*=8.2 Hz, 2H); ¹³C NMR (CDCl₃, 75 MHz): δ=51.6, 54.1, 57.3, 92.9, 130.7, 137.2, 137.8, 171.5; LR-MS (CI/NH₃): *m/z*=378 [(M+H)⁺].

4.1.3. *Z-p*-Me-Phe-O^tBu **3.** A Schlenk flask was charged with *p*-iodotoluene (0.34 g, 1.5 mmol), Pd₂(dba)₃ (40 mg, 2.5 mol%), P(*o*-tol)₃ (47 mg, 10 mol%), CuBr·DMS (32 mg, 0.1 equiv) and 4 mL of DME. At –10 °C a solution of the organozinc derivate of *Z*-I-Ala-O^tBu (1.2 mmol) in 4 mL of DME was added. The reaction mixture was slowly allowed to warm to room temperature over night. The reaction mixture was concentrated under argon and the solvent was removed in vacuum. AcOEt was added and the organic phase was washed twice with water and brine, the combined organic layers were dried over MgSO₄ and concentrated. The crude resulting oil was purified by chromatography on silica gel (hexanes/AcOEt=4:1) affording **3** (0.31 g, 66%) as a colorless solid, mp=51–52 °C. *R*_f (hexanes/AcOEt=4:1)=0.28; [α]_D²⁰+17.0 in CHCl₃; ¹H NMR (CDCl₃, 300 MHz): δ=1.41 (s, 9H), 2.31 (s, 3H), 3.04 (d, *J*=5.5 Hz, 2H), 4.51 (dt, *J*=5.9, 8.1 Hz, 1H), 5.10 (s, 2H), 5.21 (d, *J*=8.2 Hz, 1H), 7.00–7.11 (m, 4H), 7.30–7.39 (m, 5H); ¹³C NMR (CDCl₃, 75 MHz): δ=21.1, 28.0, 37.9, 52.2, 66.8, 82.3, 128.1, 128.5, 129.1, 129.4, 132.9, 136.4, 136.5, 143.4, 155.6, 170.7; LR-MS (ESI/DCM/MeOH+10 mmol/L NH₄OAc): *m/z*=756 [(2 M+NH₄)⁺], 387 [(M+NH₄)⁺], 370 [(M+H)⁺]. Anal. Calcd for C₂₂H₂₇NO₄ (369.46): C 71.52, H 7.37, N 3.79; found C 71.60, H 7.14, N 3.60.

4.1.4. *Z-p*-NH₂-Phe-O^tBu **4.** A Schlenk flask was charged with *p*-iodoaniline (0.34 g, 1.5 mmol), Pd₂(dba)₃ (40 mg, 2.5 mol%), P(*o*-tol)₃ (47 mg, 10 mol%), CuBr·DMS (32 mg, 0.1 equiv), and 4 mL of DME. At –10 °C a solution of the organozinc derivate of *Z*-I-Ala-O^tBu (1.2 mmol) in 4 mL of DME was added. The reaction mixture was slowly allowed to warm to room temperature over night. The reaction mixture was concentrated under argon and the solvent was removed under vacuum. AcOEt was added and the organic phase was washed twice with water and brine. The combined organic layers were dried over MgSO₄ and concentrated. The crude resulting oil was purified by chromatography on silica gel (3:2 hexanes/AcOEt=3:2) affording the pure product **4** (0.35 g, 73%) as a yellowish glass. *R*_f (hexanes/AcOEt=3:2)=0.2; [α]_D²⁰+40.0 in CHCl₃; ¹H NMR (CDCl₃, 300 MHz): δ=1.37

(s, 9H), 2.92 (d, $J=5.8$ Hz, 2H), 3.56 (br s, 2H), 4.41 (dt, $J=5.9, 8.1$ Hz, 1H), 5.05 (s, 2H), 5.15 (d, $J=8.0$ Hz, 1H), 6.54 (d, $J=8.2$ Hz, 2H), 6.87 (d, $J=8.2$ Hz, 2H), 7.25–7.35 (m, 5H); ^{13}C NMR (CDCl_3 , 75 MHz): $\delta=28.0, 37.5, 55.3, 66.8, 82.1, 115.2, 125.8, 128.1, 128.5, 128.6, 130.4, 136.5, 146.3, 155.7, 170.8$; LR-MS (ESI/DCM/MeOH + 10 mmol/L NH_4OAc): $m/z=388$ $[(\text{M}+\text{NH}_4)^+]$, 371 $[(\text{M}+\text{H})^+]$. Anal. Calcd for $\text{C}_{21}\text{H}_{26}\text{N}_2\text{O}_4$ (370.45): C 68.09, H 7.07, N 7.56; found C 68.23, H 7.20, N 7.38.

4.1.5. Z-p-Boc-NH-L-Phe-OMe 5. A Schlenk flask was charged with (4-iodo-phenyl)-carbamic acid *tert*-butyl ester (0.38 g, 1.2 mmol), $\text{Pd}_2(\text{dba})_3$ (40 mg, 2.5 mol%), $\text{P}(o\text{-tol})_3$ (47 mg, 10 mol%), $\text{CuBr}\cdot\text{DMS}$ (32 mg, 0.1 equiv), 2 mL of HMPT, and 2 mL of THF. At -10°C a solution of the organozinc derivate of Z-I-L-Ala-OMe (1.1 mmol) in 2 mL of THF was added. The reaction mixture was slowly allowed to warm to room temperature over night. The reaction mixture was concentrated under argon and the solvent was removed under vacuum. AcOEt was added and the organic phase was washed twice with water and brine. The collected organic layers were dried over MgSO_4 and concentrated. The crude resulting oil was purified via chromatography on silica gel (hexanes/AcOEt=2:1) affording the pure product **5** (0.39 g, 75%) as a solid, mp = 65–67 $^\circ\text{C}$. R_f (hexanes/AcOEt=2:1)=0.2; $[\alpha]_D^{20}+10.0$ in CHCl_3 ; ^1H NMR (CDCl_3 , 300 MHz): $\delta=1.51$ (s, 9H), 2.97–3.13 (m, 2H), 3.71 (s, 3H), 4.57–4.68 (m, 1H), 5.09 (s, 2H), 5.21 (d, $J=8.2$ Hz, 1H), 7.00 (d, $J=8.5$ Hz, 2H), 7.27 (d, $J=8.2$ Hz, 2H), 7.30–7.44 (m, 5H), 7.30–7.39 (m, 5H); ^{13}C NMR (CDCl_3 , 75 MHz): $\delta=28.3, 37.5, 52.4, 54.8, 67.0, 80.6, 118.6, 128.1, 128.2, 128.6, 129.9, 130.1, 136.2, 137.4, 152.7, 155.6, 172.0$; LR-MS (ESI/DCM/MeOH + 10 mmol/L NH_4OAc): $m/z=446$ $[(\text{M}+\text{NH}_4)^+]$, 429 $[(\text{M}+\text{H})^+]$, 373 $[(\text{M}+\text{H}-\text{C}_4\text{H}_8)^+]$, 329 $[(\text{M}+\text{H}-\text{Boc})^+]$. Anal. Calcd for $\text{C}_{23}\text{H}_{28}\text{N}_2\text{O}_6$ (428.49): C 64.47, H 6.59, N 6.54; found C 64.30, H 6.27, N 6.19.

4.1.6. Z-p-Me(IDA-OMe)-Phe-O^tBu 6. A Schlenk flask was charged with **9** (0.30 g, 1.2 mmol), $\text{Pd}_2(\text{dba})_3$ (40 mg, 2.5 mol%), $\text{P}(o\text{-tol})_3$ (47 mg, 10 mol%), $\text{CuBr}\cdot\text{DMS}$ (32 mg, 0.1 equiv), 2 mL of HMPT, and 2 mL of THF. At -10°C a solution of the organozinc derivate of Z-I-Ala-O^tBu (1.2 mmol) in 4 mL of THF was added. The reaction mixture was slowly allowed to warm to room temperature over night. The reaction mixture was concentrated under argon and the solvent was removed under vacuum. AcOEt was added and the organic phase was washed twice with water and brine. The collected organic layers were dried over MgSO_4 and concentrated. The crude resulting oil was purified via chromatography on silica gel (hexanes/AcOEt=7:3) affording the pure product **6** (0.41 g, 65%) as a glass. R_f (hexanes/AcOEt=7:3)=0.18; $[\alpha]_D^{20}+14.6$ in CHCl_3 ; ^1H NMR (CDCl_3 , 600 MHz): $\delta=1.42$ (s, 9H), 3.09 (d, $J=5.7$ Hz, 2H), 3.57 (s, 4H), 3.72 (s, 6H), 3.91 (s, 2H), 4.51–4.57 (m, 1H), 5.12 (s, 2H), 5.25 (d, $J=7.7$ Hz, 1H), 7.13 (d, $J=7.5$ Hz, 2H), 7.31 (d, $J=7.6$ Hz, 2H), 7.32–7.40 (m, 5H), 7.30–7.39 (m, 5H); ^{13}C NMR (CDCl_3 , 150 MHz): $\delta=27.9, 38.1, 51.5, 53.9, 55.2, 60.4, 66.8, 82.3, 128.1, 128.2, 128.5, 129.1, 129.6, 135.3, 136.0, 136.4, 155.6, 170.5, 171.5$; LR-MS (ESI/DCM/MeOH + 10 mmol/L NH_4OAc): $m/z=529$ $[(\text{M}+\text{H})^+]$, 473 $[(\text{M}+\text{H}-\text{C}_4\text{H}_8)^+]$.

Anal. Calcd for $\text{C}_{28}\text{H}_{36}\text{N}_2\text{O}_8$ (528.61): C 63.62, H 6.86, N 5.30; found C 63.79, H 6.91, N 5.20.

4.1.7. Z-p-Me(3-Boc-Cyc)-L-Phe-OMe 7. A Schlenk flask was charged with **8** (0.19 g, 0.28 mmol), $\text{Pd}_2(\text{dba})_3$ (10 mg, 2.5 mol%), $\text{P}(o\text{-tol})_3$ (12 mg, 10 mol%), $\text{CuBr}\cdot\text{DMS}$ (8 mg, 0.1 equiv), 2 mL of HMPT, and 2 mL of THF. At -10°C a solution of the organozinc derivate of Z-I-L-Ala-OMe (0.35 mmol) in 2 mL of THF was added. The reaction mixture was slowly allowed to warm to room temperature over night. The reaction mixture was concentrated under argon and the solvent was removed under vacuum. AcOEt was added and the organic phase was washed twice with water and brine. The collected organic layers were dried over MgSO_4 and concentrated. The crude resulting oil was purified via chromatography on silica gel (hexanes/AcOEt=3:2) affording the pure product **7** (0.19 g, 86%) as a solid, mp = 56–57 $^\circ\text{C}$. R_f (hexanes/AcOEt=3:2)=0.16; $[\alpha]_D^{20}+40.4$ in CHCl_3 ; ^1H NMR (CDCl_3 , 300 MHz): $\delta=1.42$ (s, 9H), 1.47 (s, 18H), 2.53–2.79 (m, 4H), 3.03–3.11 (m, 2H), 3.11–3.50 (m, 8H), 3.57 (br s, 4H), 3.69 (s, 3H), 4.57–4.70 (m, 1H), 5.08 (s, 2H), 5.21 (d, $J=7.7$ Hz, 1H), 7.02 (d, $J=8.0$ Hz, 2H), 7.16 (d, $J=8.0$ Hz, 2H), 7.30–7.44 (m, 5H), 7.30–7.38 (m, 5H); ^{13}C NMR (CDCl_3 , 75 MHz): $\delta=28.5, 28.7, 37.9, 47.3, 48.9, 49.5, 50.0, 52.3, 54.8, 56.9, 67.0, 79.3, 79.4, 128.1, 128.2, 128.6, 129.1, 130.3, 130.6, 134.7, 136.7, 155.6, 155.8, 156.1, 171.9$; HR-MS (EI/70 eV): m/z =calcd for $\text{C}_{42}\text{H}_{63}\text{N}_5\text{O}_{10}$ 797.4575; found 797.4559 ± 0.62 ppm.

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Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.tet.2005.11.034. HPLC analysis of compound **4** prepared from chiral and racemic **2a**.

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